

Vardaan Mehandiratta

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EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Applied Sciences in Computer Engineering

Present

- Coursework: C++ Programming, Microcontrollers, Calculus, Linear algebra, Classical Mechanics
- Scholarships: President's Scholarship of Distinction, David Johnston International Student Entrance Scholarship, International Student Entrance Scholarship

EXPERIENCE

Software Intern

Remote, Canada

Bharat Denim

May 2025 - Aug. 2025

- Gained hands-on exposure to the backend architecture of an inventory management system supporting 10,000+ customers and 5,000+ inventory items, under the guidance of the tech lead.
- Optimized MySQL and PostgreSQL database schemas to store inventory, orders, and product data, improving backend scalability, contributing to a 15% reduction in daily processing time.
- Collaborated with a backend team of 5 developers using Git for version control and managing codebase changes.

School Prefect and Head of Physics Department

New Delhi, Delhi

Springdales School

2024-2025

- Led the Physics Department as Head, managing and organizing over 10 academic and extracurricular events annually for a community of approximately 1000 students.

PROJECTS

UW Login Helper | Python, JavaScript, HTML, CSS

- Spearheaded development of a Chrome extension using JavaScript, HTML, and CSS to automate University of Waterloo portal login, enabling 1-click autofill for streamlined student access.
- Leveraged browser APIs to securely store and pre-fill credential data, improving user experience and reducing repetitive login steps for 100+ students.
- Implemented intuitive UI and robust error handling, resulting in increased productivity and positive feedback from the student community.

AI-Powered Netflix Recommendation Engine | React, TypeScript, Tailwind CSS, Supabase, Lovable AI

- Developed an intelligent questionnaire application using React and TypeScript that leverages AI-driven conversational flow to generate personalized Netflix recommendations based on user mood and preferences.
- Architected a context-aware AI system with custom prompt engineering and a feedback-based rating system and watched-list feature to refine results and prevent duplicate suggestions, reducing user content search and scroll time by approximately 50%.
- Deployed globally for user testing, achieving 90% accuracy in personalized recommendations with scalable Supabase backend and responsive UI.

Hospital-Induced Delirium Detection & Prevention System | Arduino, Sensors, C++, Typescript, HTML, CSS

- Engineered a delirium prevention ecosystem utilizing Arduino telemetry and custom gamified assessments, tailored across 4 distinct age demographics to ensure broad patient accessibility.
- Validated by medical professionals for its potential to reduce manual cognitive screening time by 40%, significantly streamlining ICU nursing workflows and improving response times.
- Built a secure, scalable backend to archive patient telemetry in CSV format, powering individualized dashboards for 25+ user profiles that track long-term cognitive scores and visualize performance trends for data-driven clinical review.

TECHNICAL SKILLS

Languages: Python, C++, SQL/MySQL, JavaScript, HTML/CSS, TypeScript

Frameworks: React, Node.js, Vue.js

Developer Tools: Git, VS Code, Visual Studio, PyCharm, GitHub, Microsoft Powerpoint

Libraries: pandas, NumPy, Tkinter

Core CS Knowledge: Data Structures and Algorithms, Object-Oriented Programming, Database Systems, File Handling